UNIVERSITY OF WARWICK

Paper Details

Paper Code: CS9220_B

Paper Title: High Performance Computing

Exam Period: Summer 2023

Exam Rubric

Time Allowed: 2 hours

Exam Type: Standard Examination

Approved Calculators: Permitted

Additional Stationery

Instructions

Choose to answer TWO questions from Questions 1, 2 and 3.

Question 4 is **compulsory**.

Read carefully the instructions on the answer book and make sure that the particulars required are entered on each answer book.

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The exam question paper **MUST NOT** be removed from the examination venue.

Permitted calculators: CASIO FX82, FX83 and FX85 series

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1. *This question is about the fundamental knowledge.*

(a) There are multiple different levels to achieve parallelism in a supercomputer. Discuss the 4 key ways in which parallelism can be achieved in a supercomputer, and their benefits and issues. [12]

(b) Race conditions and deadlocks can occur in any program with parallelism. What are race conditions and deadlocks, provide a pseudocode code example and how can they be avoided? [6]

(c) Control dependencies can impact the performance of some parallelisation techniques. What is a control dependency? Give a pseudocode example and describe how it can be solved. [5]

(d) The Top500 is a ranked list of the top supercomputers in the world. To rank these, a benchmark suite called LINPACK is utilised to determine the amount of Floating Point Operations (FLOP) can be calculated per second.
(i) What is a benchmark suite and why is it useful for the benchmark suite to be varied? [4]
(ii) What are the advantages and disadvantages of using a benchmark suite over a mini-application suite? [8]

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2. *This question is about parallel programming models*

(a) OpenMP is useful to perform compile-time optimisation.

(i) Describe the model OpenMP utilises. How does this differ from the model used for PThreads or Java Runnables? [4]

(ii) How does the different schedule options help OpenMP run more efficiently? [4]

(b) Message Passing Interface (MPI) sends data from one process (referred to as a rank) to another.

(i) Why is it important that MPI is an interface? [2]

(ii) Give an example on an interface specific detail and an example on what is implementation specific with regards to an `MPI_Send` instruction. [4]

(c) Both OpenMP and MPI have a reduction operation.

(i) What is a reduction operation? [3]

(ii) Describe what is achieved by **Listing 1**. [2]

(iii) Discuss, using illustrative examples if necessary, what the function in **Listing 2** does, and how the different ranks interpret the parameter in this function. [6]

(iv) What is the key differences between an OpenMP reduction and a MPI reduction [4]

Listing 1: Code example for Question 2cii

```
int validationCheck(int N, float * a) {
    int result = 1;
    #pragma omp parallel for reduction(&:result)
    for (i = 0; i < N; i++) {
        result &= (a[i] > 0.5)
    }
}
```

Listing 2: Code example for Question 2ciii

```
int MPI_Reduce(const void *sbuf, void *rbuf, int count,
    MPI_Datatype datatype, MPI_Op op, int root, MPI_Comm comm);
```

(d) MPI allows for custom data types, which allows for data which is not contiguous to be sent within a single message. One example of a custom datatype is `MPI_Type_create_indexed_block`, which takes a count, the size of the block length, an array of offsets, the old datatype and a pointer to the new datatype. What is the memory layout of `newType` in the code shown in **Listing 3**? Assume a float takes 32 bits. [6]

Listing 3: Code example for Question 2d

```
int MPI_Type_create_indexed_block(5, 2, {2, 12, 6, 16, 17},
    MPI_FLOAT, &newType);
```

3. *This question is about high-performance computing systems.*

(a) Describe the typical architecture of a cluster system, including its functional components and their roles. Explain how scalability can be achieved in a high-performance cluster system. [10]

(b) How can parity information be utilized to achieve fault tolerance in disks? [5]

(c) Explain the architectural differences between CPUs and GPUs, with a focus on their design objectives, key functional components, and the roles these components play in their respective architectures. [10]

(d) The figure below shows two types of network topologies:

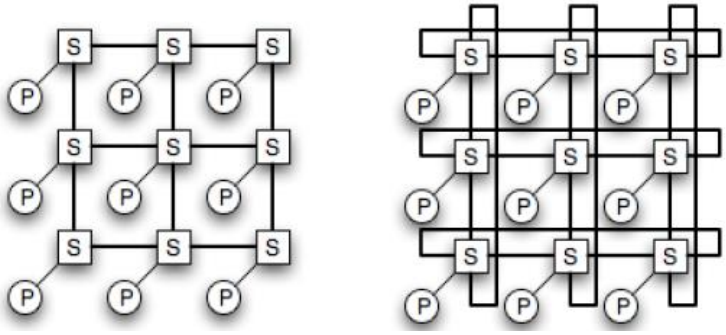

Figure 1. The topologies of two networks for Question 3(d)

i) What are the names of these two topologies? [2]

ii) Regarding the number of switches, n, in the network, what are the diameters and bisection widths of each topology? [4]

iii) Discuss which topology is superior based on their respective diameters and bisection widths. [4]

4. *This question is about performance modelling*.

(a) Assuming a linear topology, as shown in Figure 2, where switches (the squares labeled "S") use the store-and-forward switching technique to forward messages between processors (the circles labeled "P"), use the following parameters to construct a performance model for the communication time required to send a message between two processors in the linear topology.

- L: message size in bytes
- T_s: the time required to set up a communication at the sender processor
- T_h: the time of examining the head of a message in a switch or in the receiver processor
- T_w: the time of transferring a message of one byte on a network link
- D: the number of network links from the sender processor to the receive processor

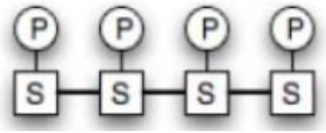

Figure 2. A linear topology for Question 4(a) [10]

(b) A program performs computations at each point in a two-dimensional grid using the 5-point stencil shown in Figure 3. The size of dimension X of the grid is N and the size of dimension Y is $4N$ (note that Figure 3 depicts only a portion of the grid), P processors are used to perform computations in parallel, assuming that N can be evenly divided by P.

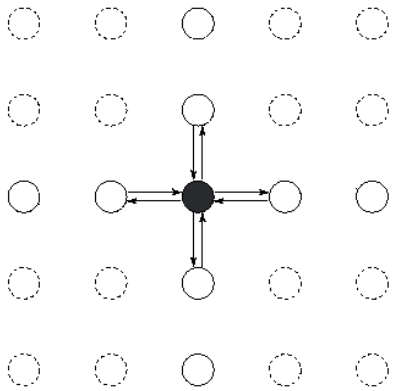

Figure 3. The communication pattern for Question 4(b)

Answer the following questions:

i) How can you construct a performance model for the program's execution time when the grid is partitioned using 1-D decomposition? Which dimension should you use to partition the grid of points, and why? Clearly define all parameters used in the constructed performance models. [5]

ii) If the grid of points is partitioned into P parts using 2-D decomposition, what is the optimal 2-D decomposition strategy in theory? Explain how you arrived at this conclusion and construct a performance model using this best 2-D decomposition strategy. Clearly define all parameters used in the constructed performance models. [8]

iii) Using the constructed performance models, derive the parallel efficiency and iso-efficiency function for both 1-D and 2-D decomposition. Which decomposition method is preferable, and why? [7]

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end